

Experiment 'p_5_3_first_digit' Results

June 4, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_5_3_first_digit

Natural language statement: Write a method returning the first digit of the argument.

Method signature: p_5_3_first_digit(n: int) returns (first: int)

Ensures

- `0 <= first < 10`
- `intToSeq(n)[0] == first`

Requires

- `0 <= n`

Functional Code Given

```
function intToSeq(n: int): seq<int>
  requires n >= 0
  ensures forall d :: d in intToSeq(n) ==> 0 <= d < 10
{
  if n < 10 then [n]
  else intToSeq(n / 10) + [n % 10]
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method returning the first digit of the argument.

The signature should be:

`p_5_3_first_digit(n: int)` returns `(first: int)`

The method should respect the following contract:

`requires 0 <= n, ensures 0 <= first < 10, ensures intToSeq(n)[0] == first`

The contract uses the following dafny code:

```
function intToSeq(n: int): seq<int>
  requires n >= 0
  ensures forall d :: d in intToSeq(n) ==> 0 <= d < 10
{
  if n < 10 then [n]
  else intToSeq(n / 10) + [n % 10]
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmyN5AcD6QQIDqE2475O3vkhcv9yf

Model: gpt-5.5-2026-04-23

Created at: 1780564183

Finish reason: stop

Usage: 468 tokens in, and 1088 tokens out

```
{
  if n < 10 {
    first := n;
  } else {
    first := p_5_3_first_digit(n / 10);
  }
}
```

Final Program

```
function intToSeq(n: int): seq<int>
  requires n >= 0
  ensures forall d :: d in intToSeq(n) ==> 0 <= d < 10
{
  if n < 10 then [n]
  else intToSeq(n / 10) + [n % 10]
}

method p_5_3_first_digit(n: int) returns (first: int)
  requires 0 <= n
  ensures 0 <= first < 10
  ensures intToSeq(n)[0] == first
{
  if n < 10 {
    first := n;
  } else {
    first := p_5_3_first_digit(n / 10);
  }
}
```

Total Token Usage

Input tokens: 468

Output tokens: 1088

Reasoning tokens: 1023

Sum of 'total tokens': 1556

Experiment Timings

Overall Experiment started at 1780564182758, ended at 1780564244618, lasting 61860ms (61.86 seconds)

Iteration #1 started at 1780564182764, ended at 1780564244618, lasting 61854ms (61.85 seconds)